



Critical Analysis Report: Olympic Insights — A Century of Performance and Athlete Profiles

CA_ONE_ (40%)

B9DA106 DATA VISUALISATION

Name: ASWIN VADUVANA BABU

Student Number: 20063083

Class: Group A (2025-26)

Programme: MSc in Data Analytics

Lecturer: Kunwar Madan

Submission Date: 10-11- 2025

1. Introduction

The Olympic Games, held since 1896, represent one of the most comprehensive and dynamic datasets in the realm of global sports analytics. Beyond athletic achievement, the Olympics reflect broader themes such as gender representation, national performance, and the evolution of physical attributes among athletes over time. This Tableau dashboard titled “Olympic Insights: A Century of Global Performance and Athlete Profiles” was developed to provide an interactive and visually coherent analysis of Olympic participation and performance from 1896 to 2016.

The visualization aims to deliver a clear narrative through five distinct yet interconnected charts: a global map of medal counts, a donut chart for seasonal distribution, a dual-axis body profile visualization, a bar chart of top-performing nations, and interactive filters that enhance exploration. Each chart was designed using the Miller Stone color palette, chosen deliberately for its subtle, earthy tones that evoke professionalism, neutrality, and accessibility.

The dashboard’s primary purpose is to enable users to answer questions such as:

- Which countries have historically dominated the Olympics?
- How do athlete physical characteristics vary by sport?
- What are the seasonal and gender-based trends in medal distribution?

The analysis below provides a detailed evaluation of design choices, data preparation, visualization methods, color theory, interactivity, and the analytical insights that emerge from the dashboard.

2. Dataset and Preprocessing

The dataset used was derived from the Kaggle Olympic Athletes Events Dataset (1896–2016), which includes over 2,70,000 records across 120 years of Olympic history. The raw dataset included multiple attributes such as athlete name, age, height, weight, sex, team, National Olympic Committee (NOC), year, season, sport, event, and medal type (Gold, Silver, Bronze, or None).

Dataset: - https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results?utm_source=chatgpt.com

2.1 Cleaning and Preparation

Substantial data cleaning was conducted before visualization to ensure consistency and accuracy:

- **Missing Value Handling:** Null values in the Medal column were replaced with the term “No Medal” for analytical clarity. This enabled seamless filtering and aggregation in Tableau without generating null-group issues.
- **Irrelevant Columns Removed:** Unnecessary or redundant columns (e.g., event-level identifiers) were excluded to streamline analysis.
- **Derived Fields:** A new ‘Decade’ column was created to group participation trends over time, improving readability for temporal charts.
- **Data Consistency:** Some NOC codes were validated and unified (e.g., USSR vs. RUS), ensuring accurate aggregation on the world map.
- **Data Type Corrections:** Numeric fields such as Age, Height, and Weight were standardized for proper averaging functions.

This cleaned dataset (`olympic_athletes_final.csv`) provided a reliable foundation for visualization, enabling efficient aggregation across key dimensions like gender, sport, season, and medal type.

3. Design Justification

The visualization design adhered to three core principles of data visualization best practice:

- **Clarity and minimalism** — ensuring zero clutter.
- **Cognitive efficiency** — designing visuals that communicate at a single glance.
- **Analytical coherence** — aligning every chart with a specific research question.

3.1 Medal Distribution by Season (Donut Chart)

Purpose: Compare total medals earned across Summer and Winter Games.

Design Rationale:

A donut chart was preferred over a standard pie chart for aesthetic and functional reasons. The central white space was used to display the total medal count (2,06,143), providing immediate quantitative context. The circular layout leverages proportional perception, with labels placed outside to prevent overlap.

Alternative visualizations such as a bar chart or tree map were tested but deemed less engaging for binary categorical comparison.

Color and Labeling Choices:

Muted tones of deep green, pale sand, and teal were used to distinguish the two seasons, maintaining harmony with the dashboard palette. Labels were set to percentage values only (e.g., Winter 16.79%), ensuring clarity without clutter. The minimalistic legend and consistent typography maintain cognitive flow.

Effectiveness:

The donut chart communicates a single insight succinctly: the Summer Games dominate medal distribution, reflecting the larger number of events compared to the Winter Olympics.

3.2 Global Map of Olympic Medal Counts

Purpose: Display global performance distribution, identifying dominant nations and participation trends.

Design Rationale:

A filled map (choropleth) was selected to intuitively represent geographical dominance using spatial context. The NOC field was assigned a geographic role (Country/Region), enabling Tableau to map medal counts accurately. A sequential Miller Stone gradient was applied (light beige to deep bronze-gold) to depict medal intensity, ensuring accessibility for colorblind viewers while avoiding harsh contrasts.

Alternative approaches considered included bubble maps and tree maps, but these introduced visual overlap and reduced geographical clarity. The chosen design achieves a clean, static representation that communicates instantly without interactivity overload. Borders, gridlines, and map controls were removed to eliminate visual noise.

Effectiveness:

The map allows immediate recognition of key powerhouses such as the USA, Russia, China, and Germany. Filtering by decade or season dynamically reveals shifting geopolitical trends (e.g., post–Cold War realignments). The simplicity of color encoding supports quick interpretation without legends dominating the view.

3.3 Average Athlete Age and Body Profile by Sport

Purpose: Examine variations in physical attributes across sports.

Design Rationale:

A dual-axis combo chart combining bars (average height) and circles (average weight) was selected. This approach allowed two quantitative measures to be compared within one coherent visual space. A sorted order by sport optimized readability, while horizontal orientation minimized label truncation.

Alternative designs, including scatter plots and box plots, were tested. However, these produced excessive overlapping points due to the density of sports data. The chosen dual-axis design-maintained clarity and balance while remaining visually compact.

Color Scheme and Formatting:

Soft gray bars and muted teal dots reflect Miller Stone's neutral tonality. Axis lines were minimized, and reference lines were added for global averages, enhancing interpretive depth without distraction.

Effectiveness:

The chart clearly highlights that sports such as Rowing and Swimming feature taller and heavier athletes, while Gymnastics and Fencing favor lighter, shorter builds. The integration of age (through the tooltip) enriches user understanding of sport-specific physiology.

3.4 Top 10 Countries by Medal Type

Purpose: Display comparative medal performance across leading nations.

Design Rationale:

A horizontal stacked bar chart was chosen for its familiarity and efficiency in showing proportional composition (Gold, Silver, Bronze). Sorting by total medal count ensures the visual hierarchy matches user expectations.

Alternative options such as tree maps were rejected due to their space inefficiency and poor readability for categorical data.

Color and Labeling: - Each medal type was encoded with a muted metallic tone (Gold #d4af37, Silver #b6b6b6, Bronze #b87333) — a natural mapping to human expectations. Labels were limited to total counts only, reducing visual noise.

Effectiveness:

At a single glance, users can identify that the USA leads in both total and gold medal counts, followed by Germany and Australia. The visual maintains proportional consistency while allowing drill-down through filters (e.g., Season or Sport).

3.5 Interactivity and Filters

The dashboard includes synchronized filters for Sport, NOC, Sex, Decade, and Medal Type. These were placed on the right-hand panel for vertical alignment and ease of access. Each filter was configured with “Apply to all worksheets using this data source”, ensuring cohesive cross-chart interaction.

Interactive Effectiveness:

Filters enable multidimensional exploration — for example, examining gender-specific trends in swimming or temporal changes in winter participation. The consistent filter styling (compact dropdowns and checkboxes) preserves space while promoting intuitive use.

Evaluation:

The interactivity was deliberately restrained — only essential filters were included to avoid “choice overload.” Hover tooltips were simplified to show key metrics, adhering to minimalism principles and maintaining interpretive focus.

3.6 Color and Label Justification

All visual elements follow a unified Miller Stone palette, consisting of soft neutrals and earthy tones (#f6f3ef, #b9925e, #8da0a6, #d8c3a5, #6b6b6b). This palette was selected to:

- Promote visual harmony and reduce cognitive strain.
- Maintain aesthetic consistency across the dashboard.
- Ensure accessibility for users with mild color-vision deficiencies.

Labels were intentionally minimal — only totals or percentages displayed — to prioritize clarity. This aligns with Tufte’s principle of maximizing the data-ink ratio, emphasizing information over decoration.

4. Key Findings and Analytical Insights

The dashboard not only visualizes Olympic data but also surfaces several meaningful patterns, correlations, and anomalies.

4.1 Global Patterns of Performance

The choropleth map reveals a distinct pattern of geographic concentration. The United States remains the dominant force, particularly in the Summer Games, followed by Russia, Germany, and China. Europe, North America, and select Asian nations collectively account for over 75% of total medals.

In contrast, regions such as Africa and South America display limited but gradually improving representation, especially post-1980s, signaling the widening inclusivity of the Olympics.

4.2 Seasonal Distribution

The donut chart shows a significant disparity: over 80% of medals originate from the Summer Olympics. This confirms that the scale of Summer events — in terms of both sports and participants — remains far greater. However, when filtered by decade, a steady growth in Winter Games diversity is evident, reflecting technological and climatic adaptation in newer host nations.

4.3 Athlete Physique Trends

The dual-axis body profile visualization uncovers sport-specific physiological patterns:

- Rowing and Swimming: Highest average height and weight, reflecting endurance and leverage requirements.
- Gymnastics and Fencing: Lowest averages, suggesting agility and precision prioritization.
- Equestrian and Sailing: Relatively balanced metrics, implying less dependency on body size.

Gender filtering further highlights that male athletes generally exceed female averages by 10–15% in both height and weight, yet female averages have steadily increased in later decades, reflecting improved training and nutrition equality.

4.4 National Dominance

The Top 10 Countries bar chart highlights systemic dominance. The USA leads all categories, followed by Germany, Australia, and Italy. Notably, countries with smaller populations (e.g., Australia) show exceptionally high medal-per-athlete ratios — a sign of training efficiency and resource focus. The visualization also

surfaces historical shifts: the USSR/Russia's peak during the Cold War era contrasts with China's rise after 2000, signaling changing global power dynamics.

4.5 Gender Representation

Using the Sex filter, gender participation trends become evident. Women's representation grows dramatically post-1970s, aligning with global gender equality movements and Olympic policy reforms. Certain sports like Gymnastics and Swimming show near parity in medal counts.

5. Conclusion

This Tableau dashboard exemplifies the synthesis of data cleaning, design thinking, and analytical interpretation. Through the lens of Olympic history, it demonstrates how visualization transforms raw data into a coherent narrative about global athletic evolution.

Each visualization type was deliberately chosen to maximize clarity and interpretive power:

- The map situates performance geographically.
- The donut chart summarizes proportionality elegantly.
- The dual-axis chart visualizes physical diversity.
- The bar chart delivers comparative ranking insight.

The dashboard's Miller Stone palette, minimalist typography, and filter interactivity collectively ensure an uncluttered, professional experience. Interactivity enhances exploration without overwhelming the user, while consistent visual language maintains cognitive flow.

From a critical standpoint, the visualization succeeds in balancing aesthetic minimalism with analytical richness. The integration of calculated ratios (e.g., medal efficiency, though not visualized directly here) offers potential for future enhancements. The design's restraint — prioritizing interpretability over novelty — ensures that insights remain accessible and academically defensible.

Ultimately, the project not only visualizes the Olympic dataset but embodies a design philosophy of clarity through simplicity, providing a reusable framework for data-driven storytelling in sports analytics and beyond.

